



Technical Description SIGMA CONTROL 2

Process Map $\geq$ booster_2.1.1

7_7601_BOOSTER_PA_03E

General

1. In communication mode "send/receive" Sigma Control can monitor the master program. This takes place with "iSlaveMsgHeader" and "iSlaveMsgFooter" that have to be returned unchanged from master.

Example sequence for Profibus Master in Step 7 :

L DB (receive box master).DBW 94	Message header
T DB (transmit box master).DBW 38	
L DB (receive box master).DBW 126	Message footer
T DB (transmit box master).DBW 62	

Monitoring can be activated in "*Communication / Com-Module / / Communication error: Timeout*" by entering a check mark at check box right of Timeout.
If monitoring is not required the above sequence is not used.

2. Availability or use of some signals depends on type of compressor, compressor options or use of functions at SC2. Information see column "Availability" or "Usability" in the following tables in combination with table 3.
For further information on signal availability and use ask your local KAESER subsidiary or KAESER Headquarter Service Support Engineering, phone +49 9561 640-7979

Sigma Control 2 Profibus

1. The data required for planning a Profibus project are in the equipment specification file "KAES0CEC.gsd".
2. The Sigma Control is a Profibus DPV0 slave with 64 bytes of input data and 128 bytes of output data. Depending on the number of Sigma Control units used the Profibus DP master must have a correspondingly large enough address space.

Sigma Control 2 Profinet

1. The data required for planning a Profinet project are in the equipment specification file "GSDML-V2.25-Kaeser-SC2-20120203.xml".

Sigma Control 2 Modbus

1. Compressor ==>>> Master Controller

Read Input Registers with modbus function 04 "Read Input Registers" or read Holding Registers with modbus function 03 "Read Holding Registers".

Modbus Protocol address: Address of the data at the bus (Base 0).

Modbus Register: Register address of the data = PLC-address (Base 1).

3xxxx **3** = Identification of Input Register, xxxx = Register address

4xxxx **4** = Identification of Holding Register, xxxx = Register address

2. Master Controller ==>>> Compressor

Write registers with function 16 "Write Multiple Registers" or 06 "Write Single Register".

Read registers with function 03 "Read Holding Registers". Never written registers contain "0".

Modbus Protocol address: Address of the data at the bus (Base 0).

Modbus Register: Register address of the data = PLC-address (Base 1).

4xxxx **4** = Identification of Holding Register, xxxx = Register address

Sigma Control 2 Devicenet

1. The data required for planning a DeviceNet project are in the equipment specification file "324-8172-EDS_ABCC_DEV_V_2_3_1.eds".
2. 8 Bit register: Low bytes are shown at low address.

SAM Send/Receive

1. Data for compressor 1 to 16 can be found in SAM data block 64 to 79.

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Table 1 Process

1. Compressor ==>>> Master Controller

1) -32768 = Value invalid or not used

2) 65535 = Value invalid or not used

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

Table 1: Process data

JSON-RPC	Sigma Control 2							SAM		Description	Availability	Unit	min	max	Format	
	Profibus/-net		Modbus**		DeviceNet	Send/Receive										
	BDSnd	S7 DBW	S5 DW	Protocol address		Input Register	Protocol address	Holding Register	Register MSB/LSB							S5 DW
	0	0	21	30022	277	40278	1/0	32	64	Alarm xAM16 - xAM1	Table 1.1	4	binary			BOOL
	2	1	22	30023	278	40279	3/2	33	66	Alarm xAM32 - xAM17	Table 1.1	4	binary			BOOL
	4	2	23	30024	279	40280	5/4	34	68	Alarm xAM48 - xAM33	Table 1.1	4	binary			BOOL
	6	3	24	30025	280	40281	7/6	35	70	Alarm xAM64 - xAM49	Table 1.1	4	binary			BOOL
	8	4	25	30026	281	40282	9/8	36	72	Alarm xAM80 - xAM65	Table 1.1	4	binary			BOOL
	10	5	26	30027	282	40283	11/10	37	74	Warning xWM16 - xWM1	Table 1.2	4	binary			BOOL
	12	6	27	30028	283	40284	13/12	38	76	Warning xWM32 - xWM17	Table 1.2	4	binary			BOOL
	14	7	28	30029	284	40285	15/14	39	78	Warning xWM48 - xWM33	Table 1.2	4	binary			BOOL
	16	8	29	30030	285	40286	17/16	40	80	Warning xWM64 - xWM49	Table 1.2	4	binary			BOOL
	18	9	30	30031	286	40287	19/18	41	82	Warning xWM80 - xWM65	Table 1.2	4	binary			BOOL
	20	10	31	30032	287	40288	21/20	42	84	Operational message xOM16 - xOM1	Table 1.3	4	binary			BOOL
	22	11	32	30033	288	40289	23/22	43	86	Operational message xOM32 - xOM17	Table 1.3	4	binary			BOOL
	24	12	33	30034	289	40290	25/24	44	88	Operational message xOM40 - xOM33	Table 1.3	4	binary			BOOL
	26	13	34	30035	290	40291	27/26	45	90	Binary signal xBS15 - xBS0	Table 1.4	4	binary			BOOL
	28	14	35	30036	291	40292	29/28	46	92	Binary signal xBS31 - xBS16	Table 1.4	4	binary			BOOL
	30	15	36	30037	292	40293	31/30	47	94	Binary signal xBS47 - xBS32	Table 1.4	4	binary			BOOL
	32	16	37	30038	293	40294	33/32	48	96	Binary signal xBS63 - xBS48	Table 1.4	4	binary			BOOL
iDV0	34	17	38	30039	294	40295	35/34	49	98	Software version 3rd digit / 4th digit		1				BYTE/BYTE
iDV1	36	18	39	30040	295	40296	37/36	50	100	Software version 1st digit / 2nd digit		1				BYTE/BYTE
iDV2	38	19	40	30041	296	40297	39/38	51	102	Pressure inlet p1		1	mbar	-400 ¹⁾	20000	INT
uiDV3	40	20	41	30042	297	40298	41/40	52	104	Pressure outlet p4		1	mbar	0	64000 ²⁾	UINT
iDV4	42	21	42	30043	298	40299	43/42	53	106	Pressure intake side p2		2.OC40	mbar	-400 ¹⁾	20000	INT
uiDV5	44	22	43	30044	299	40300	45/44	54	108	Pressure pressure side p3		2.OC40	mbar	0	64000 ²⁾	UINT
iDV6	46	23	44	30045	300	40301	47/46	55	110	Oil pressure p5		1	mbar	-1500 ¹⁾	7500	INT
iDV7	48	24	45	30046	301	40302	49/48	56	112	Block discharge temperature T1.1		1	0,1 °C	-500 ¹⁾	2999	INT
iDV8	50	25	46	30047	302	40303	51/50	57	114	Block discharge temperature T1.2		1	0,1 °C	-500 ¹⁾	2999	INT
iDV9	52	26	47	30048	303	40304	53/52	58	116	Block discharge temperature T1.3		1	0,1 °C	-500 ¹⁾	2999	INT
iDV10	54	27	48	30049	304	40305	55/54	59	118	Temperature outlet T2		1	0,1 °C	-500 ¹⁾	2999	INT
iDV11	56	28	49	30050	305	40306	57/56	60	120	Motor temperature T3		1	0,1 °C	-500 ¹⁾	2999	INT
iDV12	58	29	50	30051	306	40307	59/58	61	122	First IOM analog output		1	0,1mA	0 ¹⁾	200	INT
uiDV13	60	30	51	30052	307	40308	61/60	62	124	Pressure actual value p4 (pressure regulation)		1	mbar	0	64000 ²⁾	UINT
iDV14	62	31	52	30053	308	40309	63/62	63	126	Current drive motor starts/h		1	1			INT

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Table 1 Process

1. Compressor ==>>> Master Controller

1) -32768 = Value invalid or not used

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** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

Table 1: Process data

JSON-RPC	Sigma Control 2							SAM		Description	Availability	Unit	min	max	Format
	Profibus/-net		Modbus**		DeviceNet	Send/Receive									
	BDSnd	S7 DBW	S5 DW	Protocol address		Input Register	Protocol address	Holding Register	Register MSB/LSB						
iDV15	64	32	53	30054	309	40310	65/64	64	128	Total compressor hours run x 1000	1	h			INT
iDV16	66	33	54	30055	310	40311	67/66	65	130	Total compressor hours run x 1	1	h			INT
iDV17	68	34	55	30056	311	40312	69/68	66	132	Hours on load x 1000	1	h			INT
iDV18	70	35	56	30057	312	40313	71/70	67	134	Hours on load x 1	1	h			INT
uiDV19	72	36	57	30058	313	40314	73/72	68	136	Setpoint pressure pA switching point	1	mbar			UINT
iDV20	74	37	58	30059	314	40315	75/74	69	138	Setpoint pressure pA switching difference	1	mbar			INT
uiDV21	76	38	59	30060	315	40316	77/76	70	140	Setpoint pressure pB switching point	1	mbar			UINT
iDV22	78	39	60	30061	316	40317	79/78	71	142	Setpoint pressure pB switching difference	1	mbar			INT
uiDV23	80	40	61	30062	317	40318	81/80	72	144	Nominal package pressure	1	mbar			UINT
uiDV24	82	41	62	30063	318	40319	83/82	73	146	Increased cut-out pressure pE	1	mbar			UINT
iDV25	84	42	63	30064	319	40320	85/84	74	148	Remaining drive motor idle time	1	s			INT
iDV26	86	43	64	30065	320	40321	87/86	75	150	free					
iDV27	88	44	65	30066	321	40322	89/88	76	152	free					
iDV28	90	45	66	30067	322	40323	91/90	77	154	Frequency converter direct current link voltage	2.F	V			INT
iDV29	92	46	67	30068	323	40324	93/92	78	156	Frequency converter output current	2.F	0,1A			INT
iSlaveMsgHeader	94	47	68	30069	324	40325	95/94	79	158	Message header slave	1				UINT
iDV30	96	48	69	30070	325	40326	97/96	80	160	Frequency converter torque	2.F	0,1Nm			INT
iDV31	98	49	70	30071	326	40327	99/98	81	162	free					
iDV32	100	50	71	30072	327	40328	101/100	82	164	free					
iDV33	102	51	72	30073	328	40329	103/102	83	166	free					
iDV34	104	52	73	30074	329	40330	105/104	84	168	free					
iDV35	106	53	74	30075	330	40331	107/106	85	170	free					
iDV36	108	54	75	30076	331	40332	109/108	86	172	free					
iDV37	110	55	76	30077	332	40333	111/110	87	174	free					
iDV38	112	56	77	30078	333	40334	113/112	88	176	free					
uiDV39	114	57	78	30079	334	40335	115/114	89	178	Current cut-out pressure pA (pB)	1	mbar			UINT
iDV40	116	58	79	30080	335	40336	117/116	90	180	free					
iDV41	118	59	80	30081	336	40337	119/118	91	182	Compressor model / free	1				INT
iDV42	120	60	81	30082	337	40338	121/120	92	184	Compressor start delay	1	s			INT
iDV43	122	61	82	30083	338	40339	123/122	93	186	Duty cycle	1	1%			INT
iDV44	124	62	83	30084	339	40340	125/124	94	188	Current motor speed	2.F	1/min			INT
iSlaveMsgFooter	126	63	84	30085	340	40341	127/126	95	190	Message footer slave	1				UINT

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Table 1 Diag

1. Compressor ==>>> Master Controller

Tabelle 1: Diagnostic data

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

JSON-RPC	Sigma Control 2							DeviceNet	SAM		Description	Availability	Example		
	BDSnd	S7 DBB	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	Byte	Register MSB/LSB	S5 DW	S7 DBB		HEX	Value	Result
%	0	DL0	%	%	%	%	%	%	DL96	192	Profibus: System status 1	P			
%	1	DR0	%	%	%	%	%	%	DR96	193	Profibus: System status 2	P			
%	2	DL1	%	%	%	%	%	%	DL97	194	Profibus: System status 3	P			
%	3	DR1	%	%	%	%	%	%	DR97	195	Profibus: DP master address	P	01	1	
%	4	DL2	%	%	%	%	%	%	DL98	196	Profibus: Manufacturer code: PNO number	P	0C		Kaeser PNO Nr.: OCEC
%	5	DR2	%	%	%	%	%	%	DR98	197	Profibus: Manufacturer code: PNO number	P	EC		
%	6	DL3	%	%	%	%	%	%	DL99	198	Profibus: Header-Byte specific diagnostic data	P			
%	7	DR3	1	30002	257	40258	MSB	%	DR99	199	SC2 MCS Kaeser SN: 5th digit, 6th digit *	1*	62	62	SC2 MCS Kaeser SN: 003062
%	8	DL4	1	30002	257	40258	LSB	%	DL100	200	SC2 MCS Kaeser SN: 3rd digit, 4th digit *	1*	30	30	
%	9	DR4	2	30003	258	40259	MSB	%	DR100	201	SC2 MCS Kaeser SN: 1st digit, 2nd digit *	1*	00	00	
%	10	DL5	2	30003	258	40259	LSB	%	DL101	202	reserved		00		
%	11	DR5	8	30009	264	40265	MSB	%	DR101	203	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	SC2 MCS Kaeser PN: 7.7601P0
%	12	DL6	8	30009	264	40265	LSB	%	DL102	204	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	13	DR6	9	30010	265	40266	MSB	%	DR102	205	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	14	DL7	9	30010	265	40266	LSB	%	DL103	206	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	15	DR7	10	30011	266	40267	MSB	%	DR103	207	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	16	DL8	10	30011	266	40267	LSB	%	DL104	208	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	17	DR8	11	30012	267	40268	MSB	%	DR104	209	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	18	DL9	11	30012	267	40268	LSB	%	DL105	210	SC2 MCS Kaeser PN: space (ASCII)	1	20	SP	
%	19	DR9	4	30005	260	40261	MSB	%	DR105	211	SC2 MCS Kaeser PN: 8th digit (ASCII)	1	30	0	
%	20	DL10	4	30005	260	40261	LSB	%	DL106	212	SC2 MCS Kaeser PN: 7th digit (ASCII)	1	50	P	
%	21	DR10	5	30006	261	40262	MSB	%	DR106	213	SC2 MCS Kaeser PN: 6th digit (ASCII)	1	31	1	
%	22	DL11	5	30006	261	40262	LSB	%	DL107	214	SC2 MCS Kaeser PN: 5th digit (ASCII)	1	30	0	
%	23	DR11	6	30007	262	40263	MSB	%	DR107	215	SC2 MCS Kaeser PN: 4th digit (ASCII)	1	36	6	
%	24	DL12	6	30007	262	40263	LSB	%	DL108	216	SC2 MCS Kaeser PN: 3rd digit (ASCII)	1	37	7	
%	25	DR12	7	30008	263	40264	MSB	%	DR108	217	SC2 MCS Kaeser PN: 2nd digit (ASCII)	1	2E	.	
%	26	DL13	7	30008	263	40264	LSB	%	DL109	218	SC2 MCS Kaeser PN: 1st digit (ASCII)	1	37	7	

M Modbus only
P Profibus only
1* since manuf. date MFGDT: 2010/11
1 always

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Table 1 Diag

1. Compressor ==>>> Master Controller

Tabelle 1: Diagnostic data

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

JSON-RPC	Sigma Control 2							SAM		Description	Availability	Example			
	Profibus		Modbus **				DeviceNet	Send/Receive				HEX	Value	Result	
	BDSnd	S7 DBB	S5 DW	Input address	Input Register	Protocol address	Holding Register	Byte	Register MSB/LSB						S5 DW
%	27	DR13	12	30013	268	40269	MSB	%	DR109	219	reserved		00		
%	28	DL14	12	30013	268	40269	LSB	%	DL110	220	reserved		00		
%	29	DR14	13	30014	269	40270	MSB	%	DR110	221	reserved		00		
%	30	DL15	13	30014	269	40270	LSB	%	DL111	222	reserved		00		
%	%	%	14	30015	270	40271	MSB	%	%	%	reserved		00		
%	%	%	14	30015	270	40271	LSB	%	%	%	reserved		00		
%	%	%	15	30016	271	40272	MSB	%	%	%	reserved		00		
%	%	%	15	30016	271	40272	LSB	%	%	%	reserved		00		
%	%	%	16	30017	272	40273	MSB	%	%	%	reserved		00		
%	%	%	16	30017	272	40273	LSB	%	%	%	reserved		00		
%	%	%	17	30018	273	40274	MSB	%	%	%	reserved		00		
%	%	%	17	30018	273	40274	LSB	%	%	%	reserved		00		
%	%	%	18	30019	274	40275	MSB	%	%	%	Modbus: Controller identifier (ASCII)	M	20	SP	SC2
%	%	%	18	30019	274	40275	LSB	%	%	%	Modbus: Controller identifier (ASCII)	M	53	S	
%	%	%	19	30020	275	40276	MSB	%	%	%	Modbus: Controller identifier (ASCII)	M	43	C	
%	%	%	19	30020	275	40276	LSB	%	%	%	Modbus: Controller identifier (ASCII)	M	32	2	

M	Modbus only
P	Profibus only
1*	since manuf. date MFGDT: 2010/11
1	always

SIGMA CONTROL 2 Process Map

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Table 1.1

1. Compressor ==>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

* Summary of SC2 messages

3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always
Availability	Description / possible cause
1	Compressor drive motor overheated
1	TÜV inspection: Activating pressure of the pressure relief valve exceeded
1	EMERGENCY STOP control device actuated
1	Initial pressure too low
1	Initial pressure too high
1	A diagnostic alarm occurred
1	Permissible internal temperature for SIGMA CONTROL 2 MCS or IOM exceeded
1	The activating pressure of the pressure relief valve exceeded
1	Pressure ratio (x.x) between initial and discharge pressure too high. Factory setting x.x = 5.5
1	Door open / interlocked panel removed while the machine is running
t	1 Overload shut-down of the compressor drive motor
1	Maximum permissible airend discharge temperature (BDT) exceeded at compressed air discharge port cylinder head 1
1	Maximum permissible airend discharge temperature (BDT) exceeded at compressed air discharge port cylinder head 2
1	Maximum permissible airend discharge temperature (BDT) exceeded at compressed air discharge port cylinder head 3

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Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

* Summary of SC2 messages

3 depends on use of the function at SC 2
2.x depends on model type or options, see table 3
1 always

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**				Modbus S5	DeviceNet		SAM Send/Receive		Message		Availability	Description / possible cause
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)	
xAM17		3.0	1	22	30023	278	40279	0	2	0	33	67.0		free	
xAM18		3.1	1	22	30023	278	40279	1	2	1	33	67.1	0018	Interior fan Overcurrent	1 Overload shut-down of the interior fan motor
xAM19		3.2	1	22	30023	278	40279	2	2	2	33	67.2		free	
xAM20		3.3	1	22	30023	278	40279	3	2	3	33	67.3		free	
xAM21		3.4	1	22	30023	278	40279	4	2	4	33	67.4		free	
xAM22		3.5	1	22	30023	278	40279	5	2	5	33	67.5		free	
xAM23		3.6	1	22	30023	278	40279	6	2	6	33	67.6		free	
xAM24		3.7	1	22	30023	278	40279	7	2	7	33	67.7		free	
xAM25		2.0	1	22	30023	278	40279	8	3	0	33	66.0	*	First IOM - Group alarm	1 Fault in IOM itself
xAM26		2.1	1	22	30023	278	40279	9	3	1	33	66.1	*	Second IOM - Group alarm	2.O C40 Fault in IOM itself
xAM27		2.2	1	22	30023	278	40279	10	3	2	33	66.2		free	
xAM28		2.3	1	22	30023	278	40279	11	3	3	33	66.3		free	
xAM29		2.4	1	22	30023	278	40279	12	3	4	33	66.4	*	Third IOM - Group alarm	2.C Fault in IOM itself
xAM30		2.5	1	22	30023	278	40279	13	3	5	33	66.5		free	
xAM31		2.6	1	22	30023	278	40279	14	3	6	33	66.6		free	
xAM32		2.7	1	22	30023	278	40279	15	3	7	33	66.7		free	

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Table 1.1

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Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

* Summary of SC2 messages

3	depends on use of the function at SC 2
2.x	depends on model type or options, see table 3
1	always
Availability	Description / possible cause
	1 Oil level too low. Oil pump defective.
	2.F Mains contactor of the compressor motor does not activate
	2.F Overload shut-down of the control cabinet fan motor.
	1 Package discharge (PD) temperature too high
	2.F Mains contactor of the compressor motor does not deactivate
	1 Second power failure
	2.O C40 Machine not depressurised when motor is off. (Rest = Motor off)
2.F Fault in frequency converter for compressor motor	
2.F RS485-FC: USS bus communication error	

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** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

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1 always

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**				Modbus S5	DeviceNet		SAM Send/Receive		Message		Availability	Description / possible cause
	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX				
xAM49	7.0	3	24	30025	280	40281	0	6	0	35	71.0		free		
xAM50	7.1	3	24	30025	280	40281	1	6	1	35	71.1		free		
xAM51	7.2	3	24	30025	280	40281	2	6	2	35	71.2		free		
xAM52	7.3	3	24	30025	280	40281	3	6	3	35	71.3		free		
xAM53	7.4	3	24	30025	280	40281	4	6	4	35	71.4		free		
xAM54	7.5	3	24	30025	280	40281	5	6	5	35	71.5		free		
xAM55	7.6	3	24	30025	280	40281	6	6	6	35	71.6		free		
xAM56	7.7	3	24	30025	280	40281	7	6	7	35	71.7		free		
xAM57	6.0	3	24	30025	280	40281	8	7	0	35	70.0		free		
xAM58	6.1	3	24	30025	280	40281	9	7	1	35	70.1		free		
xAM59	6.2	3	24	30025	280	40281	10	7	2	35	70.2		free		
xAM60	6.3	3	24	30025	280	40281	11	7	3	35	70.3		free		
xAM61	6.4	3	24	30025	280	40281	12	7	4	35	70.4		free		
xAM62	6.5	3	24	30025	280	40281	13	7	5	35	70.5		free		
xAM63	6.6	3	24	30025	280	40281	14	7	6	35	70.6		free		
xAM64	6.7	3	24	30025	280	40281	15	7	7	35	70.7		free		

SIGMA CONTROL 2 Process Map

≥ booster_2.1.1

Table 1.1

1. Compressor ==>>> Master Controller

Table 1.1 : Alarms

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

* Summary of SC2 messages

3 depends on use of the function at SC 2
2.x depends on model type or options, see table 3
1 always

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**				Modbus S5	DeviceNet		SAM Send/Receive		Message		Availability	Description / possible cause
	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX				
xAM65	9.0	4	25	30026	281	40282	0	8	0	36	73.0		free		
xAM66	9.1	4	25	30026	281	40282	1	8	1	36	73.1		free		
xAM67	9.2	4	25	30026	281	40282	2	8	2	36	73.2		free		
xAM68	9.3	4	25	30026	281	40282	3	8	3	36	73.3		free		
xAM69	9.4	4	25	30026	281	40282	4	8	4	36	73.4		free		
xAM70	9.5	4	25	30026	281	40282	5	8	5	36	73.5		free		
xAM71	9.6	4	25	30026	281	40282	6	8	6	36	73.6		free		
xAM72	9.7	4	25	30026	281	40282	7	8	7	36	73.7		free		
xAM73	8.0	4	25	30026	281	40282	8	9	0	36	72.0	0073	External message 1	3	External binary message 1
xAM74	8.1	4	25	30026	281	40282	9	9	1	36	72.1	0074	External message 2	3	External binary message 2
xAM75	8.2	4	25	30026	281	40282	10	9	2	36	72.2	0075	External message 3	3	External binary message 3
xAM76	8.3	4	25	30026	281	40282	11	9	3	36	72.3	0076	External message 4	3	External binary message 4
xAM77	8.4	4	25	30026	281	40282	12	9	4	36	72.4	0077	External message 5	3	External binary message 5
xAM78	8.5	4	25	30026	281	40282	13	9	5	36	72.5	0078	External message 6	3	External binary message 6
xAM79	8.6	4	25	30026	281	40282	14	9	6	36	72.6		free		
xAM80	8.7	4	25	30026	281	40282	15	9	7	36	72.7		free		

SIGMA CONTROL 2 Process Map

≥ booster_2.1.1

Table 1.2

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

Table 1.2 : Warning Messages and Maintenance Messages												W	Warning message	3	depends on use of the function at SC 2	
												M	Maintenance message	2.x	depends on model type or options, see table 3	
												*	Summary of SC2 messages	1	always	
Sigma Control 2												SAM		Message	Availability	Description / possible cause
JSON-RPC	Profibus/-net		Modbus**				Modbus	DeviceNet		Send/Receive						
BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)			
xWM1	11.0	5	26	30027	282	40283	0	10	0	37	75.0	W 0001	Equipment number incomplete	1	The compressor equipment number consists of less than 7 characters	
xWM2	11.1	5	26	30027	282	40283	1	10	1	37	75.1	W 0002	Motor temperature T3 ↑	1	Drive motor overheating	
xWM3	11.2	5	26	30027	282	40283	2	10	2	37	75.2		free			
xWM4	11.3	5	26	30027	282	40283	3	10	3	37	75.3		free			
xWM5	11.4	5	26	30027	282	40283	4	10	4	37	75.4	W 0005	Pressure inlet p1 ↓	1	Initial pressure too low	
xWM6	11.5	5	26	30027	282	40283	5	10	5	37	75.5		free			
xWM7	11.6	5	26	30027	282	40283	6	10	6	37	75.6	W 0007	Block discharge temperature T1.1 (AIR1.03) ↑	1	Maximum permissible airend discharge temperature (BDT) reached at compressed air discharge port cylinder head 1	
xWM8	11.7	5	26	30027	282	40283	7	10	7	37	75.7	W 0008	Block discharge temperature T1.2 (AIR1.04) ↑	1	Maximum permissible airend discharge temperature (BDT) reached at compressed air discharge port cylinder head 2	
xWM9	10.0	5	26	30027	282	40283	8	11	0	37	74.0	W 0009	Block discharge temperature T1.3(AIR1.05) ↑	1	Maximum permissible airend discharge temperature (BDT) reached at compressed air discharge port cylinder head 3	
xWM10	10.1	5	26	30027	282	40283	9	11	1	37	74.1		free			
xWM11	10.2	5	26	30027	282	40283	10	11	2	37	74.2	W 0011	Target pressure ratio (pA / p1 SP = x.x) ↑	1	Target pressure ratio between target pressure pA and warning threshold for "inlet pressure p1 low" is too high. x.x = target pressure	
xWM12	10.3	5	26	30027	282	40283	11	11	3	37	74.3	W 0012	Target pressure ratio (pB / p1 SP = x.x) ↑	1	Target pressure ratio between target pressure pB and warning threshold for "inlet pressure p1 low" is too high. x.x = target pressure	
xWM13	10.4	5	26	30027	282	40283	12	11	4	37	74.4	W 0013	Target pressure ratio - Current ↑ (x.x)	1	Target pressure ratio between current target pressure and warning threshold for "inlet pressure p1 low" is too high. x.x = target pressure	
xWM14	10.5	5	26	30027	282	40283	13	11	5	37	74.5		free			
xWM15	10.6	5	26	30027	282	40283	14	11	6	37	74.6	W 0015	Bus alarm	1	The bus link from the Com-Modul interface is interrupted.	
xWM16	10.7	5	26	30027	282	40283	15	11	7	37	74.7		free			

SIGMA CONTROL 2 Process Map

≥ booster_2.1.1

Table 1.2

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

Table 1.2 : Warning Messages and Maintenance Messages												W	Warning message		3	depends on use of the function at SC 2	
** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3												M	Maintenance message		2.x	depends on model type or options, see table 3	
												*	Summary of SC2 messages		1	always	
JSON-RPC	Profibus/-net		Sigma Control 2					DeviceNet		SAM		Message		Availability	Description / possible cause		
			Modbus**				Send/Receive										
BDSnd	S7 DBX	S5 DW	Input		Holding		Modbus S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)				
xWM17	13.0	6	27	30028	283	40284	0	12	0	38	77.0	M 0017	Non-return valves h ‡	2.O C40	End of check valve service life. Replacement required.		
xWM18	13.1	6	27	30028	283	40284	1	12	1	38	77.1	M 0018	Air cooler change h ‡	1	End of cooler service life. Replacement required.		
xWM19	13.2	6	27	30028	283	40284	2	12	2	38	77.2	M 0019	Hydro damper h ‡	1	End of hydro damper service life. Replacement required.		
xWM20	13.3	6	27	30028	283	40284	3	12	3	38	77.3	M 0020	Hose lines change h ‡	1	End of hose line service life. Replacement required.		
xWM21	13.4	6	27	30028	283	40284	4	12	4	38	77.4	M 0021	Block change h ‡	1	End of compressor block service life. Replacement required.		
xWM22	13.5	6	27	30028	283	40284	5	12	5	38	77.5	M 0022	Microfilter element h ‡	1	End of micro-filter element service life. Replacement required.		
xWM23	13.6	6	27	30028	283	40284	6	12	6	38	77.6	M 0023	Dirt trap h ‡	1	Dirt trap requires servicing.		
xWM24	13.7	6	27	30028	283	40284	7	12	7	38	77.7	M 0024	Mains contactor operations ‡	2.F	Maximum switching operations of mains contactor exceeded.		
xWM25	12.0	6	27	30028	283	40284	8	13	0	38	76.0	M 0025	Belt change h ‡	1	Belt replacement required.		
xWM26	12.1	6	27	30028	283	40284	9	13	1	38	76.1	M 0026	Oil change h ‡	1	Lubricating oil: Maintenance interval has elapsed.		
xWM27	12.2	6	27	30028	283	40284	10	13	2	38	76.2	M 0027	Oil filter h ‡	1	Oil filter: Maintenance interval has elapsed.		
xWM28	12.3	6	27	30028	283	40284	11	13	3	38	76.3	M 0028	Air filter + non-return valve h ‡	2.O G40	Air filter + non-return valve: Maintenance interval has elapsed.		
xWM29	12.4	6	27	30028	283	40284	12	13	4	38	76.4	M 0029	Valve inspection h ‡	1	Valves: Maintenance interval has elapsed.		
xWM30	12.5	6	27	30028	283	40284	13	13	5	38	76.5	M 0030	Belt inspection h ‡	1	Belt tension: Maintenance interval has elapsed.		
xWM31	12.6	6	27	30028	283	40284	14	13	6	38	76.6	M 0031	Motor bearing h ‡	1	Motor bearing of compressor motor: Maintenance interval has elapsed.		
xWM32	12.7	6	27	30028	283	40284	15	13	7	38	76.7	M 0032	Electrical equipment h ‡	1	Electric components and installation: Maintenance interval has elapsed.		

SIGMA CONTROL 2 Process Map

≥ booster_2.1.1

Table 1.2

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

Table 1.2 : Warning Messages and Maintenance Messages												W	Warning message	3	depends on use of the function at SC 2
												M	Maintenance message	2.x	depends on model type or options, see table 3
												*	Summary of SC2 messages	1	always
Sigma Control 2												Message		Availability	Description / possible cause
JSON-RPC	Profibus/-net		Modbus**				DeviceNet		SAM						
			Send/Receive												
BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)		
xWM33	15.0	7	28	30029	284	40285	0	14	0	39	79.0	M 0033	Fan bearing h ‡	1	Motor bearing of fan motors: Maintenance interval has elapsed.
xWM34	15.1	7	28	30029	284	40285	1	14	1	39	79.1	M 0034	Filter mat control cabinet h ‡	1	Filter mat within the control cabinet: Maintenance interval has elapsed.
xWM35	15.2	7	28	30029	284	40285	2	14	2	39	79.2	W 0035	Temperature outlet T2 †	1	Compressed air discharge temperature T2 too high.
xWM36	15.3	7	28	30029	284	40285	3	14	3	39	79.3	W 0036	Motor starts /h ‡	1	The permissible number of motor starts was exceeded in the last 60 minutes.
xWM37	15.4	7	28	30029	284	40285	4	14	4	39	79.4	W 0037	Motor starts /d ‡	1	The permissible number of motor starts was exceeded in the last 24 hours.
xWM38	15.5	7	28	30029	284	40285	5	14	5	39	79.5	W 0038	Blow-off protection †	1	The pressure relief valve's activating pressure will soon be reached.
xWM39	15.6	7	28	30029	284	40285	6	14	6	39	79.6		free		
xWM40	15.7	7	28	30029	284	40285	7	14	7	39	79.7		free		
xWM41	14.0	7	28	30029	284	40285	8	15	0	39	78.0	W 0041	Mains voltage ↓	1	First power failure: The machine is automatically restarted.
xWM42	14.1	7	28	30029	284	40285	9	15	1	39	78.1		free		
xWM43	14.2	7	28	30029	284	40285	10	15	2	39	78.2	W 0043	External load signal?	3	External load signal indistinct: Increased cut-out pressure exceeded. The external load control has not switched to idle (off load).
xWM44	14.3	7	28	30029	284	40285	11	15	3	39	78.3		free		
xWM45	14.4	7	28	30029	284	40285	12	15	4	39	78.4	W 0045	DO test active	3	The DO test is active
xWM46	14.5	7	28	30029	284	40285	13	15	5	39	78.5	W 0046	Network pressure p4 ↓	3	Network pressure has fallen below the set 'low' value. Air consumption too high.
xWM47	14.6	7	28	30029	284	40285	14	15	6	39	78.6		free		
xWM48	14.7	7	28	30029	284	40285	15	15	7	39	78.7	M 0048	Bearing lube h ‡	1	Re-grease the motor bearings. Maintenance interval has elapsed.

SIGMA CONTROL 2 Process Map

≥ booster_2.1.1

Table 1.2

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

Table 1.2 : Warning Messages and Maintenance Messages												W M *		3 2.x 1		depends on use of the function at SC 2 depends on model type or options, see table 3 always	
Sigma Control 2												SAM		Message		Availability	Description / possible cause
JSON-RPC	Profibus/-net		Modbus**				DeviceNet	Send/Receive									
			Input		Holding			S5	S7								
BDSnd	S7 DBX	S5 DW	Protocol address	Input Register	Protocol address	Holding Register	Modbus S5 Bit			Register	Bit	DW	DBX	ID	Text (SC2 Display, except *)		
xWM49	17.0	8	29	30030	285	40286	0	16	0	40	81.0	M 0049	Annual maintenance	1	Appears one year after last maintenance.		
xWM50	17.1	8	29	30030	285	40286	1	16	1	40	81.1	M 0050	Fan bearing lube h ‡	1	Re-grease the fan bearing. Maintenance interval has elapsed.		
xWM51	17.2	8	29	30030	285	40286	2	16	2	40	81.2		free				
xWM52	17.3	8	29	30030	285	40286	3	16	3	40	81.3		free				
xWM53	17.4	8	29	30030	285	40286	4	16	4	40	81.4		free				
xWM54	17.5	8	29	30030	285	40286	5	16	5	40	81.5		free				
xWM55	17.6	8	29	30030	285	40286	6	16	6	40	81.6		free				
xWM56	17.7	8	29	30030	285	40286	7	16	7	40	81.7		free				
xWM57	16.0	8	29	30030	285	40286	8	17	0	40	80.0		free				
xWM58	16.1	8	29	30030	285	40286	9	17	1	40	80.1	W 0058	SC2 <=> SC2 Communication error	3	SC2 to SC2 communication is interrupted.		
xWM59	16.2	8	29	30030	285	40286	10	17	2	40	80.2	W 0059	Start T ↓ ↓	1	The compressor block temperature is too low (<-10°C / 14°F) for the machine to be operated.		
xWM60	16.3	8	29	30030	285	40286	11	17	3	40	80.3	W 0060	Start T ↓	1	The compressor block temperature is too low (<+2°C / 36°F).		
xWM61	16.4	8	29	30030	285	40286	12	17	4	40	80.4		free				
xWM62	16.5	8	29	30030	285	40286	13	17	5	40	80.5	W 0203	Compressor motor FC Service mode active	2.F 3	Service mode of frequency converter is active.		
xWM63	16.6	8	29	30030	285	40286	14	17	6	40	80.6	W 0240	Compressor motor FC test shut-off required	2.F 3	Test of "Safe Torque off" shut down at frequency converter.		
xWM64	16.7	8	29	30030	285	40286	15	17	7	40	80.7		free				

SIGMA CONTROL 2 Process Map

≥ booster_2.1.1

Table 1.2

1. Compressor ==>>> Master Controller

Table 1.2 : Warning Messages and Maintenance Messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

Table 1.2 : Warning Messages and Maintenance Messages												W	Warning message		3	depends on use of the function at SC 2	
												M	Maintenance message		2.x	depends on model type or options, see table 3	
												*	Summary of SC2 messages		1	always	
JSON-RPC	Profibus/-net		Sigma Control 2					Modbus	DeviceNet		SAM		Message		Availability	Description / possible cause	
			Modbus**		Send/Receive												
BDSnd	S7 DBX	S5 DW	Input		Holding		S5 Bit	Register	Bit	S5 DW	S7 DBX	ID	Text (SC2 Display, except *)				
			Protocol address	Input Register	Protocol address	Holding Register											
xWM65	19.0	9	30	30031	286	40287	0	18	0	41	83.0		free				
xWM66	19.1	9	30	30031	286	40287	1	18	1	41	83.1	0066	free				
xWM67	19.2	9	30	30031	286	40287	2	18	2	41	83.2	0067	free				
xWM68	19.3	9	30	30031	286	40287	3	18	3	41	83.3	W 0068	Condensate drain	2.1	The condensate drain is defective.		
xWM69	19.4	9	30	30031	286	40287	4	18	4	41	83.4	0069	free				
xWM70	19.5	9	30	30031	286	40287	5	18	5	41	83.5	W 0070	Oil pressure p5 ↑	1	Oil pressure too high.		
xWM71	19.6	9	30	30031	286	40287	6	18	6	41	83.6	W 0071	Oil level ↓	1	Oil level too low.		
xWM72	19.7	9	30	30031	286	40287	7	18	7	41	83.7	0069	free				
xWM73	18.0	9	30	30031	286	40287	8	19	0	41	82.0	W 0073	External message 1	3	External binary message 1		
xWM74	18.1	9	30	30031	286	40287	9	19	1	41	82.1	W 0074	External message 2	3	External binary message 2		
xWM75	18.2	9	30	30031	286	40287	10	19	2	41	82.2	W 0075	External message 3	3	External binary message 3		
xWM76	18.3	9	30	30031	286	40287	11	19	3	41	82.3	W 0076	External message 4	3	External binary message 4		
xWM77	18.4	9	30	30031	286	40287	12	19	4	41	82.4	W 0077	External message 5	3	External binary message 5		
xWM78	18.5	9	30	30031	286	40287	13	19	5	41	82.5	W 0078	External message 6	3	External binary message 6		
xWM79	18.6	9	30	30031	286	40287	14	19	6	41	82.6		free				
xWM80	18.7	9	30	30031	286	40287	15	19	7	41	82.7		free				

SIGMA CONTROL 2 Process Map

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Table 1.3

1. Compressor ==>>> Master Controller

Table 1.3 : Operational messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

* Summary of SC2 messages

3 depends on use of the function at SC 2
2.x depends on model type or options, see table 3
1 always

Sigma Control 2											SAM		Summary of SC2 messages		Availability	Description / possible cause
JSON-RPC	Profibus/-net		Modbus**				Modbus	DeviceNet		Send/Receive		Message				
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address				Holding Register	S5 Bit			S5 DW		
													ID	Text (SC2 Display, except *)		
xOM1	21.0	10	31	30032	287	40288	0	20	0	42	85.0			free		
xOM2	21.1	10	31	30032	287	40288	1	20	1	42	85.1			free		
xOM3	21.2	10	31	30032	287	40288	2	20	2	42	85.2			free		
xOM4	21.3	10	31	30032	287	40288	3	20	3	42	85.3			free		
xOM5	21.4	10	31	30032	287	40288	4	20	4	42	85.4	0005	Pressure inlet p1 ↓	1	Initial pressure is low	
xOM6	21.5	10	31	30032	287	40288	5	20	5	42	85.5			free		
xOM7	21.6	10	31	30032	287	40288	6	20	6	42	85.6			free		
xOM8	21.7	10	31	30032	287	40288	7	20	7	42	85.7			free		
xOM9	20.0	10	31	30032	287	40288	8	21	0	42	84.0	0009	Compressor on	1	The machine is switched on. Motor is running or compressor is ready for operation.	
xOM10	20.1	10	31	30032	287	40288	9	21	1	42	84.1	0010	Controller on	1	The controller is switched on with the «ON» key	
xOM11	20.2	10	31	30032	287	40288	10	21	2	42	84.2	0011	Cold start release	3	Cold start is enabled	
xOM12	20.3	10	31	30032	287	40288	11	21	3	42	84.3			free		
xOM13	20.4	10	31	30032	287	40288	12	21	4	42	84.4			free		
xOM14	20.5	10	31	30032	287	40288	13	21	5	42	84.5			free		
xOM15	20.6	10	31	30032	287	40288	14	21	6	42	84.6			free		
xOM16	20.7	10	31	30032	287	40288	15	21	7	42	84.7			free		

SIGMA CONTROL 2 Process Map

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Table 1.3

1. Compressor ==>>> Master Controller

Table 1.3 : Operational messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

* Summary of SC2 messages

JSON-RPC	Sigma Control 2						Modbus		DeviceNet		SAM		Message		Availability	Description / possible cause
	BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX				
xOM17		23.0	11	32	30033	288	40289	0	22	0	43	87.0	0073	External message 1	3	External binary message 1
xOM18		23.1	11	32	30033	288	40289	1	22	1	43	87.1	0074	External message 2	3	External binary message 2
xOM19		23.2	11	32	30033	288	40289	2	22	2	43	87.2	0075	External message 3	3	External binary message 3
xOM20		23.3	11	32	30033	288	40289	3	22	3	43	87.3	0076	External message 4	3	External binary message 4
xOM21		23.4	11	32	30033	288	40289	4	22	4	43	87.4	0077	External message 5	3	External binary message 5
xOM22		23.5	11	32	30033	288	40289	5	22	5	43	87.5	0078	External message 6	3	External binary message 6
xOM23		23.6	11	32	30033	288	40289	6	22	6	43	87.6		free		
xOM24		23.7	11	32	30033	288	40289	7	22	7	43	87.7		free		
xOM25		22.0	11	32	30033	288	40289	8	23	0	43	86.0		free		
xOM26		22.1	11	32	30033	288	40289	9	23	1	43	86.1		free		
xOM27		22.2	11	32	30033	288	40289	10	23	2	43	86.2	0027	Power OFF → ON	1	Request: Switch the power supply off and on.
xOM28		22.3	11	32	30033	288	40289	11	23	3	43	86.3	0028	Dynamic motor T ↑	1	Control mode DYNAMIC: The temperature of the compressor motor is too high.
xOM29		22.4	11	32	30033	288	40289	12	23	4	43	86.4		free		
xOM30		22.5	11	32	30033	288	40289	13	23	5	43	86.5	0030	Voltage restored	1	Mains power supply returned after mains failure. SC2 has been feed with supply voltage.
xOM31		22.6	11	32	30033	288	40289	14	23	6	43	86.6		free		
xOM32		22.7	11	32	30033	288	40289	15	23	7	43	86.7		free		

SIGMA CONTROL 2 Process Map

≥ booster_2.1.1

Table 1.3

1. Compressor ==>>> Master Controller

Table 1.3 : Operational messages

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

* Summary of SC2 messages

JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**				Modbus S5	DeviceNet		SAM Send/Receive		Message		Availability	Description / possible cause
	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX				
xOM33	25.0	12	33	30034	289	40290	0	24	0	44	89.0	0033	Machine report	3	Life sign has been sent.
xOM34	25.1	12	33	30034	289	40290	1	24	1	44	89.1		free		
xOM35	25.2	12	33	30034	289	40290	2	24	2	44	89.2		free		
xOM36	25.3	12	33	30034	289	40290	3	24	3	44	89.3		free		
xOM37	25.4	12	33	30034	289	40290	4	24	4	44	89.4		free		
xOM38	25.5	12	33	30034	289	40290	5	24	5	44	89.5	*	Safety shutdown ...	1	Safety shutdown by safety part of SC2 software occurred.
xOM39	25.6	12	33	30034	289	40290	6	24	6	44	89.6		free		
xOM40	25.7	12	33	30034	289	40290	7	24	7	44	89.7	*	IOSlot Undervoltage error	1	Undervoltage error at at least one IOM.

3 depends on use of the function at SC 2
2.x depends on model type or options, see table 3
1 always

1. Compressor ==>>>> Master Controller

Table 1.4 : Binary Signals

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

Table 1.4 : Binary Signals													3	depends on use of the function		
													2.x	see table 3		
** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3													1	always		
JSON-RPC	Profibus/-net		Sigma Control 2					Modbus	DeviceNet		SAM		Description	Availability	Signal	
			Modbus**		Send/Receive											
BDSnd	S7 DBX	S5 DW	Input		Holding		S5 Bit	Register	Bit	S5 DW	S7 DBX					
Protocol address	Input Register	Protocol address	Holding Register													
xBS0	27.0	13	34	30035	290	40291	0	26	0	45	91.0	Controller on (LED at key "I" on) + Remote on + Bus (Receive)	1	yes	no	
xBS1	27.1	13	34	30035	290	40291	1	26	1	45	91.1	Motor running	1	yes	no	
xBS2	27.2	13	34	30035	290	40291	2	26	2	45	91.2	Load run	1	yes	no	
xBS3	27.3	13	34	30035	290	40291	3	26	3	45	91.3	Idle	1	yes	no	
xBS4	27.4	13	34	30035	290	40291	4	26	4	45	91.4	Local mode (LED at key "Remote" off)	1	yes	no	
xBS5	27.5	13	34	30035	290	40291	5	26	5	45	91.5	Controller on (LED at key "I" on)	1	yes	no	
xBS6	27.6	13	34	30035	290	40291	6	26	6	45	91.6	Group alarm	1	yes	no	
xBS7	27.7	13	34	30035	290	40291	7	26	7	45	91.7	Group warning or group maintenance	1	yes	no	
xBS8	26.0	13	34	30035	290	40291	8	27	0	45	90.0	Compressor ready for load	1	yes	no	
xBS9	26.1	13	34	30035	290	40291	9	27	1	45	90.1	Compressor ON (Motor is running or ready for operation)	1	yes	no	
xBS10	26.2	13	34	30035	290	40291	10	27	2	45	90.2					
xBS11	26.3	13	34	30035	290	40291	11	27	3	45	90.3					
xBS12	26.4	13	34	30035	290	40291	12	27	4	45	90.4					
xBS13	26.5	13	34	30035	290	40291	13	27	5	45	90.5	Interior fan is running	1	yes	no	
xBS14	26.6	13	34	30035	290	40291	14	27	6	45	90.6	Group maintenance (maintenance messages only)	1	yes	no	
xBS15	26.7	13	34	30035	290	40291	15	27	7	45	90.7	Group warning (warning messages only)	1	yes	no	

3 depends on use of the function
 2.x see table 3
 1 always

1. Compressor ==>>> Master Controller

Table 1.4 : Binary Signals

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

Table 1.4 : Binary Signals													3	depends on use of the function		
													2.x	see table 3		
** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3													1	always		
JSON-RPC	Profibus/-net		Sigma Control 2					Modbus	DeviceNet		SAM		Description	Availability	Signal	
			Modbus**		Send/Receive											
BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX				"1"	"0"
xBS16	29.0	14	35	30036	291	40292	0	28	0	46	93.0	Mode compressor ON from "I" key only is active	3	yes	no	
xBS17	29.1	14	35	30036	291	40292	1	28	1	46	93.1	Mode compressor ON from "I" key and clock is active	3	yes	no	
xBS18	29.2	14	35	30036	291	40292	2	28	2	46	93.2					
xBS19	29.3	14	35	30036	291	40292	3	28	3	46	93.3					
xBS20	29.4	14	35	30036	291	40292	4	28	4	46	93.4	Mode compressor ON from "I" key and external contact is active	3	yes	no	
xBS21	29.5	14	35	30036	291	40292	5	28	5	46	93.5	Mode compressor ON from "I" key and clock or external contact is active	3	yes	no	
xBS22	29.6	14	35	30036	291	40292	6	28	6	46	93.6	Mode compressor ON from "I" key and bus master is active	3	yes	no	
xBS23	29.7	14	35	30036	291	40292	7	28	7	46	93.7	Mode compressor ON (xBS16..xBS22) set from	3	bus master	local	
xBS24	28.0	14	35	30036	291	40292	8	29	0	46	92.0	DUAL control mode is active	3	yes	no	
xBS25	28.1	14	35	30036	291	40292	9	29	1	46	92.1	QUADRO control mode is active	3	yes	no	
xBS26	28.2	14	35	30036	291	40292	10	29	2	46	92.2	VARIO control mode is active	3	yes	no	
xBS27	28.3	14	35	30036	291	40292	11	29	3	46	92.3	DYNAMIC control mode is active	3	yes	no	
xBS28	28.4	14	35	30036	291	40292	12	29	4	46	92.4	Continuous control mode is active	3	yes	no	
xBS29	28.5	14	35	30036	291	40292	13	29	5	46	92.5					
xBS30	28.6	14	35	30036	291	40292	14	29	6	46	92.6					
xBS31	28.7	14	35	30036	291	40292	15	29	7	46	92.7	Control mode (xBS24 - xBS28) set from	3	bus master	local	

3 depends on use of the function
2.x see table 3
1 always

1. Compressor ==>>> Master Controller

Table 1.4 : Binary Signals

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

Table 1.4 : Binary Signals													Availability	3 depends on use of the function		
														2.x	see table 3	
														1	always	
** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3													Description	Signal		
Sigma Control 2																
Modbus**																
JSON-RPC	Profibus/-net		Input				Modbus	DeviceNet		Send/Receive						
			Holding		S5					S7						
BDSnd	S7 DBX	S5 DW	Protocol address	Input Register	Protocol address	Holding Register	Bit	Register	Bit	DW	DBX		"1"	"0"		
xBS32	31.0	15	36	30037	292	40293	0	30	0	47	95.0					
xBS33	31.1	15	36	30037	292	40293	1	30	1	47	95.1					
xBS34	31.2	15	36	30037	292	40293	2	30	2	47	95.2					
xBS35	31.3	15	36	30037	292	40293	3	30	3	47	95.3					
xBS36	31.4	15	36	30037	292	40293	4	30	4	47	95.4	Mode load-idle signal from external contact is active	3	yes	no	
xBS37	31.5	15	36	30037	292	40293	5	30	5	47	95.5					
xBS38	31.6	15	36	30037	292	40293	6	30	6	47	95.6	Mode load-idle signal from bus master is active	3	yes	no	
xBS39	31.7	15	36	30037	292	40293	7	30	7	47	95.7	Mode load-idle signal (xBS32..xBS47) set from	3	bus master	local	
xBS40	30.0	15	36	30037	292	40293	8	31	0	47	94.0	Mode load-idle signal from pressure regualtor, pA permanent is active	3	yes	no	
xBS41	30.1	15	36	30037	292	40293	9	31	1	47	94.1	Mode load-idle signal from pressure regualtor, pB permanent is active	3	yes	no	
xBS42	30.2	15	36	30037	292	40293	10	31	2	47	94.2	Mode load-idle signal from pressure regualtor, pA/pB change via clock is active	3	yes	no	
xBS43	30.3	15	36	30037	292	40293	11	31	3	47	94.3	Mode load-idle signal from pressure regualtor, pA/pB change via timer is active	3	yes	no	
xBS44	30.4	15	36	30037	292	40293	12	31	4	47	94.4					
xBS45	30.5	15	36	30037	292	40293	13	31	5	47	94.5	Mode load-idle signal from pressure regualtor, pA/pB change via external contact is active	3	yes	no	
xBS46	30.6	15	36	30037	292	40293	14	31	6	47	94.6	Mode load-idle signal from pressure regualtor, pA/pB change via SC2 master (Ethernet) is active	3	yes	no	
xBS47	30.7	15	36	30037	292	40293	15	31	7	47	94.7					

3 depends on use of the function
 2.x see table 3
 1 always

1. Compressor ==>>>> Master Controller

Table 1.4 : Binary Signals

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

Table 1.4 : Binary Signals													3	depends on use of the function	
													2.x	see table 3	
													1	always	
** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3													Availability	Signal	
JSON-RPC	Profibus/-net		Sigma Control 2 Modbus**				Modbus	DeviceNet		SAM		Description			
			Send/Receive												
BDSnd	S7 DBX	S5 DW	Input Protocol address	Input Register	Holding Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX			"1"	"0"
xBS48	33.0	16	37	30038	293	40294	0	32	0	48	97.0				
xBS49	33.1	16	37	30038	293	40294	1	32	1	48	97.1				
xBS50	33.2	16	37	30038	293	40294	2	32	2	48	97.2				
xBS51	33.3	16	37	30038	293	40294	3	32	3	48	97.3	Automatic restart after power on	1	activated	deactivated
xBS52	33.4	16	37	30038	293	40294	4	32	4	48	97.4				
xBS53	33.5	16	37	30038	293	40294	5	32	5	48	97.5				
xBS54	33.6	16	37	30038	293	40294	6	32	6	48	97.6				
xBS55	33.7	16	37	30038	293	40294	7	32	7	48	97.7	Message acknowledgement by bus master	1	activated	deactivated
xBS56	32.0	16	37	30038	293	40294	8	33	0	48	96.0				
xBS57	32.1	16	37	30038	293	40294	9	33	1	48	96.1				
xBS58	32.2	16	37	30038	293	40294	10	33	2	48	96.2				
xBS59	32.3	16	37	30038	293	40294	11	33	3	48	96.3				
xBS60	32.4	16	37	30038	293	40294	12	33	4	48	96.4				
xBS61	32.5	16	37	30038	293	40294	13	33	5	48	96.5				
xBS62	32.6	16	37	30038	293	40294	14	33	6	48	96.6				
xBS63	32.7	16	37	30038	293	40294	15	33	7	48	96.7				

3 depends on use of the function
2.x see table 3
1 always

SIGMA CONTROL 2 Process Map

≥ booster_2.1.1

Table 2 Process

2. Master Controller ==>>> Compressor

* negative value

1) write both values at a time

2) write both values at a time

Table 2: Process data

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

3 see associated table (2.x)
2.x see table 3
1 always

JSON-RPC BDRcv	Sigma Control 2				DeviceNet Register MSB/LSB	SAM Send/Receive		Description	Usability	Unit	Format
	S7 DBW	S5 DW	Protocol address	Holding Register		S5 DW	S7 DBW				
	0	0	0	40001	1/0	0	0	Receive bit xBS15 - xBS0	3	binary	BOOL
	2	1	1	40002	3/2	1	2	Receive bit xBS31 - xBS16	3	binary	BOOL
	4	2	2	40003	5/4	2	4	Receive bit xBS47 - xBS32	3	binary	BOOL
	6	3	3	40004	7/6	3	6	Receive bit xBS63 - xBS48	3	binary	BOOL
uiDV0	8	4	4	40005	9/8	4	8	Setpoint pressure pA switching point ¹⁾	1	mbar	UINT
iDV1	10	5	5	40006	11/10	5	10	Setpoint pressure pA switching difference * ¹⁾	1	mbar	INT
uiDV2	12	6	6	40007	13/12	6	12	Setpoint pressure pB switching point ²⁾	1	mbar	UINT
iDV3	14	7	7	40008	15/14	7	14	Setpoint pressure pB switching difference * ²⁾	1	mbar	INT
iDV4	16	8	8	40009	17/16	8	16	Speed	2.F	n ⁻¹	INT
iDV5	18	9	9	40010	19/18	9	18				
iDV6	20	10	10	40011	21/20	10	20				
iDV7	22	11	11	40012	23/22	11	22				
iDV8	24	12	12	40013	25/24	12	24				
uiDV9	26	13	13	40014	27/26	13	26	Actual air receiver system pressure	2.F	mbar	UINT
iDV10	28	14	14	40015	29/28	14	28				
iDV11	30	15	15	40016	31/30	15	30				
iDV12	32	16	16	40017	33/32	16	32				
	34	17	17	40018	35/34	17	34	Receive bit xBS79 - xBS64	3	binary	BOOL
	36	18	18	40019	37/36	18	36	Receive bit xBS95 - xBS80	3	binary	BOOL
iMasterMsgHeader	38	19	19	40020	39/38	19	38	Message header master	1		UINT
iDV13	40	20	20	40021	41/40	20	40				
iDV14	42	21	21	40022	43/42	21	42				
iDV15	44	22	22	40023	45/44	22	44				
iDV16	46	23	23	40024	47/46	23	46				
iDV17	48	24	24	40025	49/48	24	48				
iDV18	50	25	25	40026	51/50	25	50				
iDV19	52	26	26	40027	53/52	26	52				
iDV20	54	27	27	40028	55/54	27	54				
iDV21	56	28	28	40029	57/56	28	56				
iDV22	58	29	29	40030	59/58	29	58				
iDV23	60	30	30	40031	61/60	30	60				
iMasterMsgFooter	62	31	31	40032	63/62	31	62	Message footer master	1		UINT

SIGMA CONTROL 2 Process Map

≥ booster_2.1.1

Table 2 Buspar

2. Master Controller ==>>>> Compressor

Table 2: Busparameter, Modbus only

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

* not for Modbus TCP

Sigma Control 2							SAM		Description	Example		
JSON-RPC BDSnd	Profibus/Profinet S7 DBB	S5 DW	Modbus ** Protocol address	Holding Register	Bit	DeviceNet Register MSB/LSB	Send/Receive S5 DW	S7 DBB		HEX	Value	Result
%	%	%	512	40513		%	%	%	Modbus: Slaveaddress	64	100	Slave: 100
%	%	%	513	40514		%	%	%	Modbus: Priority, stop bits, baudrate *	16		19200/even/1stop
									Even parity, 1 stop bit (default)			
									Odd parity, 1 stop bit			
									No parity, 2 stop bits			
									No parity, 1 stop bit			
			513	40514	0				0 1 0 1			
			513	40514	1				0 0 1 1			
									1200bit/s			
									2400bit/s			
									4800bit/s			
									9600bit/s			
									19200bit/s (default)			
									38400bit/s			
									57600bit/s			
									76800bit/s			
									115200bit/s			
			513	40514	2				0 1 0 1 0 1 0 1 0			
			513	40514	3				0 0 1 1 0 0 1 1 0			
			513	40514	4				0 0 0 0 1 1 1 1 0			
			513	40514	5				0 0 0 0 0 0 0 0 1			
			513	40514	6				reserved, set to 0			
			513	40514	7				reserved, set to 0			
%	%	%	514	40515		%	%	%	Modbus: Modus: 0=RTU, 1=ASCII *	0	RTU	Modus: RTU
%	%	%	515	40516		%	%	%	Modbus: Timeout in ms	7D0	2000	Timeout: 2000ms

2. Master Controller ==>>> Compressor

Table 2.1.: Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

JSON-RPC	Sigma Control 2		Modbus**		Modbus S5 Bit	DeviceNet		SAM Send/Receive		Description	Usability 2.x 1	Signal		
	BDRcv	S7 DBX	S5 DW	Protocol address	Holding Register	Register	Bit	S5 DW	S7 DBX			"1"	"0"	Kind
xBS0		1.0	0	0	40001	0	0	0	1.0	SAM connection	1	yes	no	duration
xBS1		1.1	0	0	40001	1	0	0	1.1	Load run by bus master (if xBS2 = 1)	1	load run	idle	duration
xBS2		1.2	0	0	40001	2	0	0	1.2	Mode load-idle signal from bus master	1	yes (load/idle xBS1)	local	duration
xBS3		1.3	0	0	40001	3	0	0	1.3	Continue drive motor running (if xBS0=1)	1	continue running	no bearing	duration
xBS4		1.4	0	0	40001	4	0	0	1.4					
xBS5		1.5	0	0	40001	5	0	0	1.5					
xBS6		1.6	0	0	40001	6	0	0	1.6					
xBS7		1.7	0	0	40001	7	0	0	1.7					
xBS8		0.0	0	0	40001	8	1	0	0.0					
xBS9		0.1	0	0	40001	9	1	0	0.1					
xBS10		0.2	0	0	40001	10	1	0	0.2					
xBS11		0.3	0	0	40001	11	1	0	0.3					
xBS12		0.4	0	0	40001	12	1	0	0.4					
xBS13		0.5	0	0	40001	13	1	0	0.5					
xBS14		0.6	0	0	40001	14	1	0	0.6					
xBS15		0.7	0	0	40001	15	1	0	0.7					

2. Master Controller ==>>>> Compressor

Table 2.1.: Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

JSON-RPC	Sigma Control 2		Modbus**		Modbus S5 Bit	DeviceNet		SAM Send/Receive		Description	Usability 2.x 1 depends on model type or options, see table 3 always	Signal		
	BDRcv	S7 DBX	S5 DW	Protocol address	Holding Register	Register	Bit	S5 DW	S7 DBX			"1"	"0"	Kind
xBS16		3.0	1	1	40002	0	2	0	1	3.0	1	yes	local	duration
xBS17		3.1	1	1	40002	1	2	1	1	3.1	1	yes	local	duration
xBS18		3.2	1	1	40002	2	2	2	1	3.2				
xBS19		3.3	1	1	40002	3	2	3	1	3.3				
xBS20		3.4	1	1	40002	4	2	4	1	3.4	1	yes	local	duration
xBS21		3.5	1	1	40002	5	2	5	1	3.5	1	yes	local	duration
xBS22		3.6	1	1	40002	6	2	6	1	3.6	1	yes (ON/OFF via xBS23)	local	duration
xBS23		3.7	1	1	40002	7	2	7	1	3.7	1	ON	OFF	duration
xBS24		2.0	1	1	40002	8	3	0	1	2.0	1	DUAL	local if xBS24 and xBS25 and xBS26 and xBS27 and xBS28 is "0"	duration
xBS25		2.1	1	1	40002	9	3	1	1	2.1	1	QUADRO		duration
xBS26		2.2	1	1	40002	10	3	2	1	2.2	1	VARIO		duration
xBS27		2.3	1	1	40002	11	3	3	1	2.3	1	DYNAMIC		duration
xBS28		2.4	1	1	40002	12	3	4	1	2.4	1	Continuous		duration
xBS29		2.5	1	1	40002	13	3	5	1	2.5				
xBS30		2.6	1	1	40002	14	3	6	1	2.6				
xBS31		2.7	1	1	40002	15	3	7	1	2.7				

2. Master Controller ==>>>> Compressor

Table 2.1.: Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3											2.x 1	depends on model type or options, see table 3 always		
Sigma Control 2										Usability	Signal			
JSON-RPC	Profibus/Profinet		Modbus**		Modbus	DeviceNet		SAM Send/Receive						Description
BDRcv	S7 DBX	S5 DW	Protocol address	Holding Register	S5 Bit	Register	Bit	S5 DW	S7 DBX		"1"	"0"	Kind	
xBS32	5.0	2	2	40003	0	4	0	2	5.0					
xBS33	5.1	2	2	40003	1	4	1	2	5.1					
xBS34	5.2	2	2	40003	2	4	2	2	5.2					
xBS35	5.3	2	2	40003	3	4	3	2	5.3					
xBS36	5.4	2	2	40003	4	4	4	2	5.4	Mode <i>load-idle signal from external contact</i>	1	yes	local	duration
xBS37	5.5	2	2	40003	5	4	5	2	5.5					
xBS38	5.6	2	2	40003	6	4	6	2	5.6	Mode <i>load-idle signal from bus master</i>	1	yes (load/idle via xBS39)	local	duration
xBS39	5.7	2	2	40003	7	4	7	2	5.7	<i>Load run</i> by bus master (if xBS38 = 1)	1	load run	idle	duration
xBS40	4.0	2	2	40003	8	5	0	2	4.0	Mode <i>load-idle signal from pressure regualtor, pA permanent</i>	1	yes	local	duration
xBS41	4.1	2	2	40003	9	5	1	2	4.1	Mode <i>load-idle signal from pressure regualtor, pB permanent</i>	1	yes	local	duration
xBS42	4.2	2	2	40003	10	5	2	2	4.2	Mode <i>load-idle signal from pressure regualtor, pA/pB change via clock</i>	1	yes	local	duration
xBS43	4.3	2	2	40003	11	5	3	2	4.3	Mode <i>load-idle signal from pressure regualtor, pA/pB change via timer</i>	1	yes	local	duration
xBS44	4.4	2	2	40003	12	5	4	2	4.4					
xBS45	4.5	2	2	40003	13	5	5	2	4.5	Mode <i>load-idle signal from pressure regualtor, pA/pB change via external contact</i>	1	yes	local	duration
xBS46	4.6	2	2	40003	14	5	6	2	4.6	Mode <i>load-idle signal from pressure regualtor, pA/pB change via SC2 master (Ethernet)</i>	1	yes	local	duration
xBS47	4.7	2	2	40003	15	5	7	2	4.7					

2.x depends on model type or options, see table 3
1 always

2. Master Controller ==>>>> Compressor

Table 2.1.: Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3											2.x	depends on model type or options, see table 3		
											1	always		
Sigma Control 2											Usability	Signal		
JSON-RPC	Profibus/Profinet		Modbus**		Modbus S5 Bit	DeviceNet		SAM Send/Receive		Description		"1"	"0"	Kind
BDRcv	S7 DBX	S5 DW	Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX					
xBS48	7.0	3	3	40004	0	6	0	3	7.0	Set speed	2.F	bus master (iDV4)	local	duration
xBS49	7.1	3	3	40004	1	6	1	3	7.1					
xBS50	7.2	3	3	40004	2	6	2	3	7.2					
xBS51	7.3	3	3	40004	3	6	3	3	7.3					
xBS52	7.4	3	3	40004	4	6	4	3	7.4					
xBS53	7.5	3	3	40004	5	6	5	3	7.5					
xBS54	7.6	3	3	40004	6	6	6	3	7.6	Enable message acknowledgement by bus master	1	yes (ack. via xBS55)	local	duration
xBS55	7.7	3	3	40004	7	6	7	3	7.7	Acknowledge signal for alarms and warnings (if xBS54=1)	1	L→H: acknowledge		flank
xBS56	6.0	3	3	40004	8	7	0	3	6.0	Enable compressor autostart by bus master	1	yes (set via xBS57)	local	duration
xBS57	6.1	3	3	40004	9	7	1	3	6.1	Compressor restart (if xBS56=1)	1	yes	no	duration
xBS58	6.2	3	3	40004	10	7	2	3	6.2					
xBS59	6.3	3	3	40004	11	7	3	3	6.3					
xBS60	6.4	3	3	40004	12	7	4	3	6.4					
xBS61	6.5	3	3	40004	13	7	5	3	6.5					
xBS62	6.6	3	3	40004	14	7	6	3	6.6					
xBS63	6.7	3	3	40004	15	7	7	3	6.7					

2. Master Controller ==>>>> Compressor

Table 2.1.: Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3											2.x	depends on model type or options, see table 3		
Sigma Control 2											1	always		
JSON-RPC	Profibus/Profinet		Modbus**		Modbus S5 Bit	DeviceNet		SAM Send/Receive		Description	Usability	Signal		
BDRcv	S7 DBX	S5 DW	Protocol address	Holding Register		Register	Bit	S5 DW	S7 DBX			"1"	"0"	Kind
xBS64	35.0	17	17	40018	0	34	0	17	35.0					
xBS65	35.1	17	17	40018	1	34	1	17	35.1					
xBS66	35.2	17	17	40018	2	34	2	17	35.2					
xBS67	35.3	17	17	40018	3	34	3	17	35.3					
xBS68	35.4	17	17	40018	4	34	4	17	35.4	Set pressure pA (switching point and switching difference) via	1	bus master (iDV0+iDV1)	local	duration
xBS69	35.5	17	17	40018	5	34	5	17	35.5					
xBS70	35.6	17	17	40018	6	34	6	17	35.6	Set pressure pB (switching point and switching difference) via	1	bus master (iDV2+iDV3)	local	duration
xBS71	35.7	17	17	40018	7	34	7	17	35.7					
xBS72	34.0	17	17	40018	8	35	0	17	34.0					
xBS73	34.1	17	17	40018	9	35	1	17	34.1					
xBS74	34.2	17	17	40018	10	35	2	17	34.2					
xBS75	34.3	17	17	40018	11	35	3	17	34.3					
xBS76	34.4	17	17	40018	12	35	4	17	34.4					
xBS77	34.5	17	17	40018	13	35	5	17	34.5	Set actual air receiver system pressure via	2.F	bus master (iDV9)	local	duration
xBS78	34.6	17	17	40018	14	35	6	17	34.6					
xBS79	34.7	17	17	40018	15	35	7	17	34.7					

2.x depends on model type or options, see table 3
1 always

2. Master Controller ==>>>> Compressor

Table 2.1.: Receive bit assignment

** Modbus: Please see paragraph "Sigma Control 2 Modbus" at page 3

JSON-RPC	Sigma Control 2		Modbus**		Modbus S5 Bit	DeviceNet		SAM Send/Receive		Description	Usability 2.x 1	Signal		
	BDRcv	S7 DBX	S5 DW	Protocol address	Holding Register	Register	Bit	S5 DW	S7 DBX			"1"	"0"	Kind
xBS80		37.0	18	18	40019	0	36	0	18	37.0				
xBS81		37.1	18	18	40019	1	36	1	18	37.1				
xBS82		37.2	18	18	40019	2	36	2	18	37.2				
xBS83		37.3	18	18	40019	3	36	3	18	37.3				
xBS84		37.4	18	18	40019	4	36	4	18	37.4				
xBS85		37.5	18	18	40019	5	36	5	18	37.5				
xBS86		37.6	18	18	40019	6	36	6	18	37.6				
xBS87		37.7	18	18	40019	7	36	7	18	37.7				
xBS88		36.0	18	18	40019	8	37	0	18	36.0				
xBS89		36.1	18	18	40019	9	37	1	18	36.1				
xBS90		36.2	18	18	40019	10	37	2	18	36.2				
xBS91		36.3	18	18	40019	11	37	3	18	36.3				
xBS92		36.4	18	18	40019	12	37	4	18	36.4				
xBS93		36.5	18	18	40019	13	37	5	18	36.5				
xBS94		36.6	18	18	40019	14	37	6	18	36.6				
xBS95		36.7	18	18	40019	15	37	7	18	36.7				

depends on model type or options, see table 3
always

Table 3: Availability of signals for Standard machines

This table covers standard machines. In case of doubt see wiring diagram of the machine.

Avalability	CNC	DNC	Description
2.1	●	x	Condensate drain
2.F	SFC machines only		
2.F	all machines, except SFC machines		
2.O Xyy	machines with option Xyy* only , see wiring diagram of the machine		
2.O Xyy	all machines, except machines with option Xyy*, see wiring diagram of the machine		
2.C	C-machine, see wiring diagram of the machine		

x = not available

● = available

* Option

C40 Nitrogen version